accounts for weird things like virtual particles that can quantum-fluctuate in and out of existence. But calculations of the vacuum energy required for quantum field theory give us a number 120 orders of magnitude larger than what the cosmological constant actually out there in space seems to be. In case you're not familiar with the term, an order of magnitude is a factor of 10. 100 is two orders of magnitude. 120 orders of magnitude is 10 raised to the power of 120. Even in astrophysics, where we sometimes play fast and loose with the numbers, this looks like a major discrepancy. So, if the cosmological constant isn't the vacuum energy quantum field theorists all know and love, what is it?

One suggested solution to this "cosmological constant problem" involves the hypothesis that the constant is small in our observable universe, but might take other values far away, and it's just a matter of chance that we are where we are. (Or, not chance, but necessity, if vastly different values of the cosmological constant would be hostile in some way to the development of life and intelligence, perhaps by making space expand too fast for galaxies to form.) Another possibility is that it's not a cosmological constant at all, but some kind of new cosmological-constant-mimicking energy field in the universe that might change over time, in which case there's a possibility that it evolved to what it is for some other reason.

Because we don't know whether it's really a cosmological constant or not, we generally call any hypothesized phenomenon that could make the universe accelerate in its expansion *dark energy*. To throw some more terminology into the mix, an evolving (i.e., nonconstant) dark energy is often called *quintessence*, after the "fifth element," a mysterious something-or-other that was popular to philosophize about in the Middle Ages and is not really much more precisely specified now. A nice thing about the quintessence hypothesis is that it could lead us to a theory with some parallels to the cosmic inflation at the beginning of time. We know that whatever it was that caused cosmic inflation eventually turned off, so perhaps a similar accelerated-expansion-causing field could have turned on since then, causing the acceleration we see today.

(One downside of the quintessence hypothesis is that it's theoretically possible for a dark energy that changes over time to violently destroy the universe. For instance, if whatever is accelerating the expansion now turns around, it could cause the universe to stop and recollapse, bringing us back